

# Symmetric Cryptography Fundamentals

## Introduction

Symmetric cryptography, also known as **secret-key cryptography**, is one of the oldest and most widely used encryption techniques. It uses a **single shared key** for both encryption and decryption.

Because the same key is used by both the sender and receiver, the key must be exchanged securely before communication begins.

Symmetric cryptography is significantly **faster** than asymmetric cryptography, making it the preferred choice for encrypting large amounts of data.

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## Learning Objectives

After reading this note, you should understand:

- What symmetric cryptography is
  - Information theory fundamentals
  - Entropy
  - Hamming Distance
  - Hamming Weight
  - Claude Shannon's contribution to cryptography
  - Confusion and Diffusion
  - Avalanche Effect
  - Kerckhoffs's Principle
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## What is Symmetric Cryptography?

Symmetric cryptography is an encryption technique where the **same secret key** is used for both encryption and decryption.

### Encryption

Ciphertext = Encrypt(Plaintext, Secret Key)

## Decryption

$\text{Plaintext} = \text{Decrypt}(\text{Ciphertext}, \text{Secret Key})$

Mathematically,

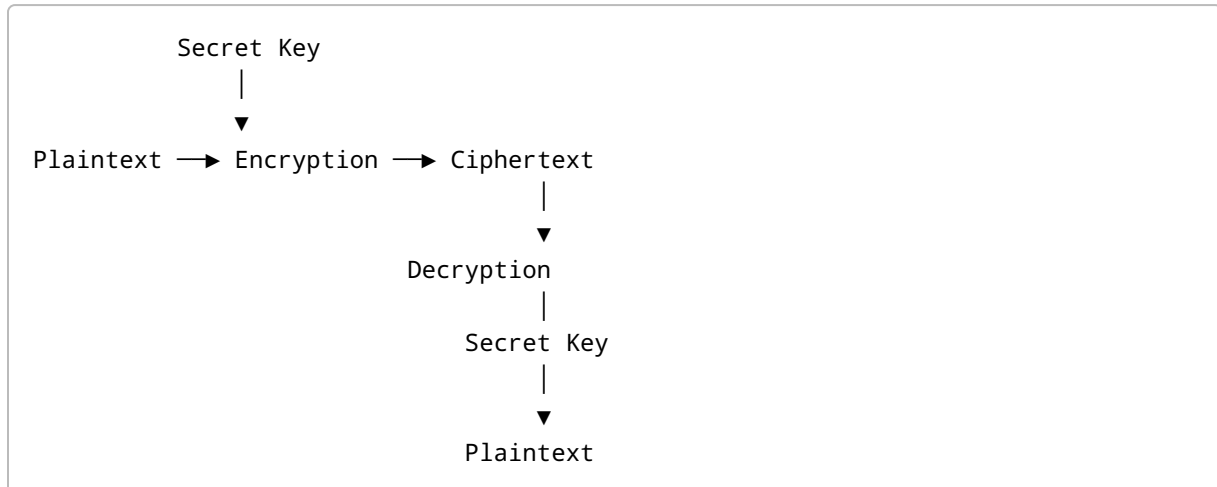
$C = E(K, P)$

$P = D(K, C)$

Where:

- **P** = Plaintext
- **C** = Ciphertext
- **K** = Secret Key
- **E()** = Encryption Function
- **D()** = Decryption Function

## How Symmetric Encryption Works



Both parties **must possess the same secret key**.

## Advantages

- Extremely fast
- Efficient for encrypting large files
- Requires less computational power

- Widely supported in hardware and software
  - Used in almost every secure communication protocol
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## Disadvantages

- Secure key exchange is required.
  - Poor scalability (each pair of users needs a unique key).
  - If the key is compromised, all encrypted data can be decrypted.
  - Does not provide non-repudiation.
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## Where is Symmetric Encryption Used?

Symmetric cryptography is widely used in:

- HTTPS (after the TLS handshake)
  - VPNs
  - SSH
  - WPA2/WPA3 Wi-Fi security
  - BitLocker
  - VeraCrypt
  - ZIP and 7z encryption
  - Database encryption
  - Full disk encryption
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## Information Theory

Information Theory is the mathematical study of **information, communication, and uncertainty**.

Modern cryptography is heavily based on information theory.

The field was pioneered by **Claude Shannon**, often called the **Father of Information Theory**.

In 1949, Shannon published the landmark paper:

*Communication Theory of Secrecy Systems*

This paper laid the mathematical foundation for modern cryptography.

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# Entropy

Entropy measures the **amount of uncertainty or randomness** in a message.

Higher entropy means:

- Less predictable data
- Stronger randomness
- Better cryptographic security

Lower entropy means:

- More predictable data
- Easier to guess
- Weaker security

The entropy of a random variable is defined as:

$$H(X) = -\sum P_i \log_2(P_i)$$

Where:

- **H(X)** = Entropy
- **P<sub>i</sub>** = Probability of each event

## Example

Random password:

P@9xL#2q

High entropy.

Weak password:

password123

Low entropy.

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# Hamming Distance

The **Hamming Distance** is the number of positions where two equal-length strings differ.

Example:

```
10110101
10011101
```

Different bits:

```
00101000
```

Hamming Distance = 2

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## Why is Hamming Distance Important?

It is used in:

- Error detection
  - Error correction
  - Cryptography
  - Coding theory
  - Digital communications
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## Hamming Weight

Hamming Weight is the **number of 1s in a binary value**.

Example:

```
10110110
```

Contains five **1s**.

Therefore,

Hamming Weight = 5

It is also known as:

- Population Count
- Popcount

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## Claude Shannon

Claude Shannon is widely regarded as the **Father of Modern Cryptography**.

His contributions include:

- Information Theory
- Entropy
- Confusion
- Diffusion
- Mathematical foundations of secure communication

Many modern encryption algorithms are based on Shannon's principles.

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## Shannon's Principles

Modern symmetric encryption relies on two important concepts:

- Confusion
- Diffusion

Together they make encrypted data resistant to cryptanalysis.

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## Confusion

Confusion hides the relationship between:

- Plaintext
- Ciphertext
- Encryption key

The goal is to make it extremely difficult for an attacker to derive the secret key from the ciphertext.

Confusion is mainly achieved using:

- Substitution
  - S-Boxes
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## Diffusion

Diffusion spreads the influence of a single plaintext bit across many ciphertext bits.

Changing one plaintext character should affect many output bits.

This makes statistical attacks much more difficult.

Diffusion is mainly achieved using:

- Permutations
  - P-Boxes
  - Multiple encryption rounds
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## Avalanche Effect

A secure encryption algorithm should exhibit a strong **Avalanche Effect**.

A tiny change in the plaintext or key should produce a **completely different ciphertext**.

### Example

Plaintext:

HELLO

Encrypted:

9F8C1A...

Now change one letter:

HELLo

Encrypted:

3A2E7F...

Although only one character changed, the ciphertext is almost entirely different.

A good cipher aims for approximately **50% of the output bits** to change when one input bit changes.

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## Kerckhoffs's Principle

In 1883, **Auguste Kerckhoffs** proposed one of the most important principles in cryptography:

*A cryptosystem should remain secure even if everything about the system is public knowledge except the secret key.*

This means:

- The algorithm can be public.
- The implementation can be public.
- The attacker may know exactly how the encryption works.
- **Only the key must remain secret.**

Modern encryption algorithms such as AES, RSA, and ChaCha20 all follow this principle.

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## Security Through Obscurity

Keeping an encryption algorithm secret instead of relying on strong mathematics is called **Security Through Obscurity**.

This approach is **not considered secure**.

History has shown that once a secret algorithm becomes public, it is often broken quickly.

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## Real-World Example

When you visit an HTTPS website:

1. TLS uses **asymmetric cryptography** to securely exchange a session key.
2. After the key exchange, **symmetric encryption (usually AES)** encrypts all subsequent communication.



This combines the strengths of both cryptographic systems.

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## Key Takeaways

- Symmetric cryptography uses one shared secret key.
  - It is much faster than asymmetric encryption.
  - Entropy measures randomness.
  - Hamming Distance counts differing bits.
  - Hamming Weight counts the number of 1s.
  - Claude Shannon introduced the foundations of modern cryptography.
  - Confusion hides relationships between keys and ciphertext.
  - Diffusion spreads plaintext changes throughout the ciphertext.
  - A good cipher exhibits a strong Avalanche Effect.
  - Kerckhoffs's Principle states that security should depend only on the secrecy of the key, not the algorithm.
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## Further Reading

- Claude Shannon – *Communication Theory of Secrecy Systems*
- NIST Cryptographic Standards
- FIPS 197 (Advanced Encryption Standard)
- EC-Council Certified Encryption Specialist (ECES)
- Bruce Schneier – *Applied Cryptography*